

Logbook 5 Template (for offline log preparation)

Name:	Choo Zhi Yi
Week Date:	Part B -Meeting 5 of 8

*(The above information is for you to keep track of which week this log is applicable for. This information is not required in the online logbook as the system will find your name, your supervisor's name, and take the submission date, automatically. You must submit the online log in the relevant week. Logs **cannot** be backdated to missed weeks.)*

Meeting date: <i>(please note if no meeting is planned this week)</i>	23 September 2026 (Wednesday)
Work Undertaken this week: <i>Not more than 500 words</i>	State the work done from last meeting in Part A (after submission Project Outline) till now? - Setup and update Raspberry Pi 5 - Setup Pi Connect between the Pi 5 and Laptop - Create a GitHub Repository Take pictures of your item prepared to date. Update your BOM Table  No changes in BOM list since Part A Hardware: - Raspberry Pi 5 8GB (Device, Power Plug, Connection Wire, Case) - 2D LiDAR Camera - 2-Wheeled Robot Chassis - Motor Driver L298N - Motors & Wheels Set

	- Wheel Encoders - 3.7V Rechargeable Batteries - LEDs - Resistors (330, 1000, 10000 Ohms) - Push Button Switches - Bolts & Nuts Sets Software installed on your laptop - Piconnect - GitHub - GitHub into Raspberry Pi Device via Piconnect Your Github link: https://github.com/clymzy/IMP_Delivery																																																																																																																																																																																				
Problems encountered: <i>(Not more than 500 words</i>	What are the challenges to your project Outline that you have made or going to make? Why? Whether I can complete the project within the 3 months given. Whether I can follow the initial Gantt Chart plans on time and in order.																																																																																																																																																																																				
Tasks agreed for upcoming week before the next meeting	What are your plan actions/task until the next progress meeting that are agreed with your supervisor - Setup Motors and Wheel Encoders - Setup LiDAR and do Autonomous Navigation																																																																																																																																																																																				
Figures: <i>(Upload image files in the online Log Form, but feel free to keep notes here about which images you intend to upload. Use "Figure Caption" information as the filenames, e.g., "Figure 1 – A screenshot of the initial constrained Sketch in CATIA.jpg")</i>	You can upload to Github and share the link No need to post pictures or videos here I updated my Gantt chart to reflect what has been done so far, the snippet features weeks 1-2: <table><tr><th rowspan="2">WBS NUMBER</th><th rowspan="2">TASK TITLE</th><th rowspan="2">START DATE</th><th rowspan="2">DUE DATE</th><th rowspan="2">DURATION</th><th rowspan="2">PCT OF TASK COMPLETE</th><th colspan="4">WEEK 0 (6/9/2026-12/9/2026)</th><th colspan="4">WEEK 1 (13/9/2026-19/9/2026)</th><th colspan="4">WEEK 2 (20/9/2026-26/9/2026)</th><th colspan="4">WEEK 3 (27/9/2026-3/10/2026)</th></tr><tr><th>M</th><th>T</th><th>W</th><th>R</th><th>F</th><th>M</th><th>T</th><th>W</th><th>R</th><th>F</th><th>M</th><th>T</th><th>W</th><th>R</th><th>F</th><th>M</th><th>T</th><th>W</th><th>R</th><th>F</th></tr><tr><td>1</td><td>Order Hardware</td><td>9/6/26</td><td>9/14/26</td><td>8</td><td>100%</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td>Download Software needed</td><td>9/15/26</td><td>9/18/26</td><td>3</td><td>100%</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td>Gather and Assemble Hardware</td><td>9/15/26</td><td>9/25/26</td><td>10</td><td>50%</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td>Code Motor Control</td><td>9/17/26</td><td>9/20/26</td><td>3</td><td>0%</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td>Testing Motor Control w/ Hardware Integration</td><td>9/20/26</td><td>9/23/26</td><td>3</td><td>0%</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>6</td><td>Code Wheel Encoder</td><td>9/23/26</td><td>9/26/26</td><td>3</td><td>0%</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table> 1) Update Hardware Block diagram of your system 2) Key software codes/modules that need to be developed 3) Any Quality and Regulatory issues addressed? 4) An updated/improved version of your system sketch My initial robot sketch did not include any storage compartment to store the items that the robot must deliver, so I had to rework the robot sketch to include it.	WBS NUMBER	TASK TITLE	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE	WEEK 0 (6/9/2026-12/9/2026)				WEEK 1 (13/9/2026-19/9/2026)				WEEK 2 (20/9/2026-26/9/2026)				WEEK 3 (27/9/2026-3/10/2026)				M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	M	T	W	R	F	1	Order Hardware	9/6/26	9/14/26	8	100%																		2	Download Software needed	9/15/26	9/18/26	3	100%																		3	Gather and Assemble Hardware	9/15/26	9/25/26	10	50%																		4	Code Motor Control	9/17/26	9/20/26	3	0%																		5	Testing Motor Control w/ Hardware Integration	9/20/26	9/23/26	3	0%																		6	Code Wheel Encoder	9/23/26	9/26/26	3	0%																	
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https://github.com/clymzy/IMP_Delivery/blob/main/IMG_1292.jpeg

I had added a storage compartment, tray-shaped with a ledge to keep the items inside.

When the robot arrives at the location it will pop up slowly to present the delivered items to the receiver (maybe a switch can be added to allow the sender to choose whether the storage should pop out or not, especially if the item is fragile). The storage should be able to fit A4 documents or a normal 500ml PET bottle drink, the storage should be made of a lightweight material to not load the wheels + chassis too much.

The LiDAR camera has to be placed on a structure taller than the delivery compartment when popped out, the LiDAR camera requires an unobstructed view of the surroundings to do SLAM properly.

I also added an ultrasensor near the LiDAR camera area, this sensor will check that the area near the camera is clear before the robot can move safely.

As the storage pop-out idea will take up some space and requires more height, the LiDAR camera is placed at a very vulnerable height and is prone to breakage as a very fragile camera.